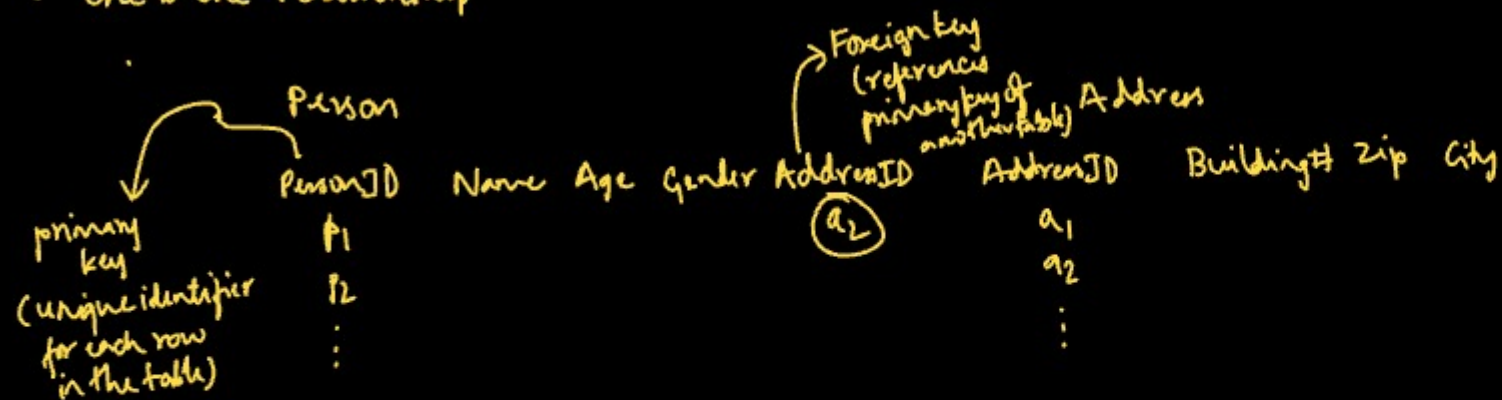
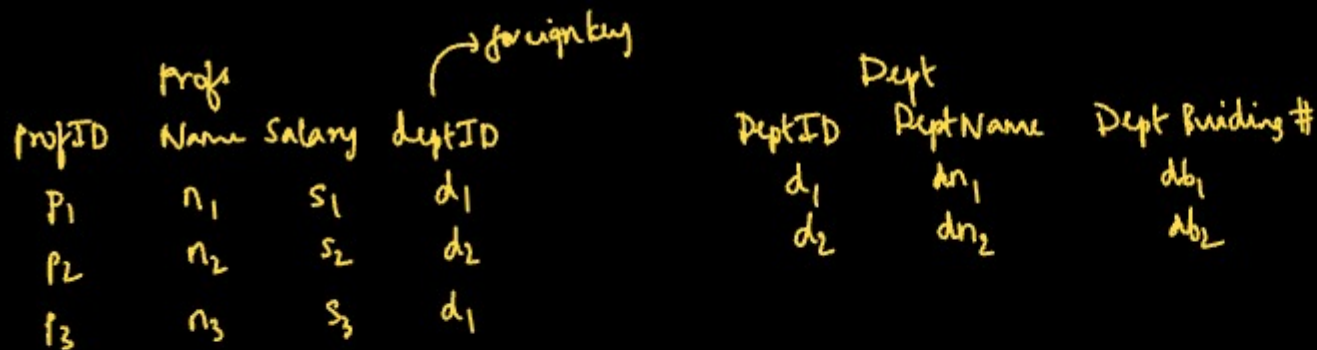


One to one relationship



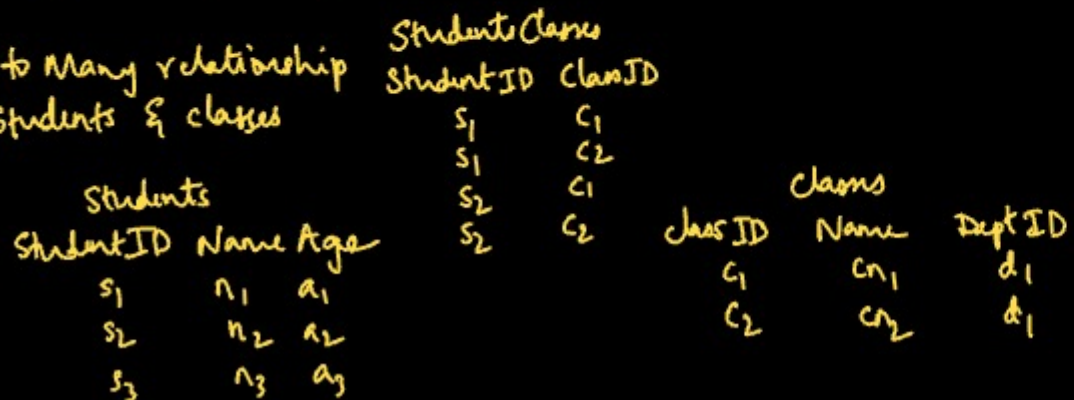
One to many relationship

- Each professor belongs to exactly one dept
- Each dept can have multiple professors



Many to Many relationship

- Students & classes



8 bits = 1 byte

Kilo-1000

2^{10} Bytes = 1024 Bytes $\sim 10^3$ Bytes = 1 KB

$2^{10} \sim 10^3$

2^{20} Bytes: $\sim 10^6$ Bytes = 1 MB

$2^{20} \sim 10^6$

2^{30} Bytes: $\sim 10^9$ Bytes = 1 GB

Mega- 10^6

2^{40} Bytes: $\sim 10^{12}$ Bytes = 1 TB

Giga- 10^9

2^{50} Bytes: $\sim 10^{15}$ Bytes = 1 PB

$$(a^5)^2 = a^{10}$$

$$a^5 \times a^5 = a^{10}$$

$$(a^m)^n = a^{mn}$$

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

D B A M A
 $S_5 | S_4 | S_3 | S_2 | S_1$ Stages

Pipelining

- divide instructions into stages
- serve multiple instrns at the same time (different stages)
- increases throughput

Cache

Registers - 32 registers

L1 Cache (each 64bit/32bit)

L2 Cache

L3 Cache

Main Memory (RAM)

Disk

Cost $\uparrow$
Capacity $\downarrow$
Speed $\uparrow$

$a++$ } critical section

1. load a from mem into reg.
2. inc reg
3. store the value in memory

T_1
1000 $\rightarrow$ load a
inc reg

store

T_2

load $a \rightarrow 1000$
inc reg
store

1 ns

10^0

100 ns

10^2

10^5

Cache:

- Temporal locality
- Spatial locality

L_1 cache 1ns
 Branch misprediction 3ns
 L_2 cache ref 4ns
 mutex lock/unlock 17ns

Main memory ref 100ns

Compressing 1KB - 2000ns = 2 μ s

SSD read - 16000ns = 16 μ s
(random)

Read 10⁶ Bytes = 1MB = 3000ns = 3 μ s
sequentially from memory

Round trip in the same data = 500,000ns = 500 μ s
center

Reading 1MB sequentially = 49000ns = 49 μ s
from SSD

Disk seek: 2000000ns = 2000 μ s = 2ms

packet roundtrip CA to Ned = 150ms
 = 150000 μ s
 = 150000000ns

1GHz $\rightarrow$ 10⁹ clock cycles/second

$\downarrow$
 Hertz - 1/second
 frequency

Giga - 10⁹

$\downarrow$
 $\Rightarrow$ Each clock
 cycle takes 1/10⁹ sec

10⁹ - Giga

10⁶ - Mega

10³ - kilo

10⁻³ - milli

10⁻⁶ - micro

10⁻⁹ - nano

Availability Numbers:

• 4 nines $\Rightarrow$ 99.99%

• 6 nines $\Rightarrow$ 99.9999%

1 Day = 1440 min
 = 86400s

Availability	Downtime per day	Downtime per year
99%	14.4 min	3.65 days
$\vdots$	$\vdots$	$\vdots$

Assumptions:

- 300 million monthly active users
- 50% of users daily
- 2 tweets per day
- 10% of tweets contain media
- Data is stored for 5 years

Queries per second

- 150 million user per day
- 300 million tweets per day
- $\frac{300 \times 10^6}{86400} \approx 3500$ QPS

$$= \frac{3000}{864} \times 1000$$

$$\approx 3.5 \times 1000$$

$$\frac{99987}{9.1} = \frac{100000}{10}$$

$$\frac{0.987 \times 10^5}{1} = \frac{1 \times 10^5}{1}$$

$$864 \times 3 \approx 2600$$

$$\text{Peak QPS} = 2 \times 3500 \text{ QPS} = 7000 \text{ QPS}$$

Media storage:	tweet ID	64 Bytes
	text	140 Bytes
	media	1 MB

$$\text{storage per day: } 30 \text{ million tweets have media} = 30 \text{ million} \times 1 \text{ MB} = 3 \times 10^7 \times 10^6 \text{ Bytes}$$

$$= 3 \times 10^{13} \text{ bytes} = 30 \text{ TB}$$

$$\text{Across 5 years: } 30 \text{ TB} \times 5 \times 365 = 3 \times 365 = 1100$$

$$30 \times 365 = 11000$$

$$\times 5 = 55000$$

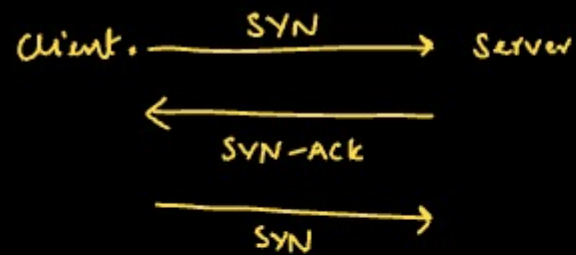
$$= 55000 \text{ TB}$$

$$= 55 \text{ PB}$$

• Layered model

- * Application $\sim$ HTTP
Presentation
Session
- * Transport $\sim$ TCP, UDP, QUIC
- * Network layer $\sim$ IP
Data link layer
Physical layer

• TCP 3way handshake



• Encryption

- Symmetric key encryption
 - encrypt & decrypt with the same secret key
 - fast
- Asymmetric key encryption
 - public-private key
 - encrypt with public key
 - decrypt with corresponding private key
 - slower
- Secret key (symmetric) shared using asymmetric key encryption scheme

• Network layer

$2^{10} \sim 10^3$

• IP

• Two versions of IP addresses

- IPv4 - 4 byte $\rightarrow 2^{32} \sim 4 \times 10^9$ (4 Billion)
- IPv6 - 16 byte

private
& public IP

- 10.0.0.0/8
- 192.168.0.0/16

private addresses

- 18.0.0.0/8 $\rightarrow$ # hosts we can have in this network = 2^{24}

$\hookrightarrow$ represents the # bits to represent the network

$\Rightarrow$ 24 bits - represent the host on the network

$$2^8 = 256$$

0 to 255

Network address translator



• Transport layer

- 4 tuple $\rightarrow$ src IP
src port
dest IP
dest port

• TCP

- Connection oriented
- Reliable delivery
- Flow control
- Congestion control
- 20-60 bytes

• UDP

- Connectionless
- No guarantee of delivery
- No ordering
- Lower latency
- 8 bytes header

- Application layer protocols

- HTTP / HTTPS

- REST: Architectural style

- Resources (Nouns)

- Methods (verbs): GET, POST, PUT, PATCH, DELETE

- /users/{id} GET

- /users POST

- /users/{id}/posts GET

method

headers: key-value pairs for metadata

can request for a certain compressed format but server might not respond with that format
- backward compatibility

Response:

status code

2xx - successful
4xx - client side error
5xx - server side error

200 OK
201 Created
404 Not found
401 Unauthorized
403
500
502

- GraphQL

- RPCs

- userService.updateData(..)

- gRPC

- Protobufs

```

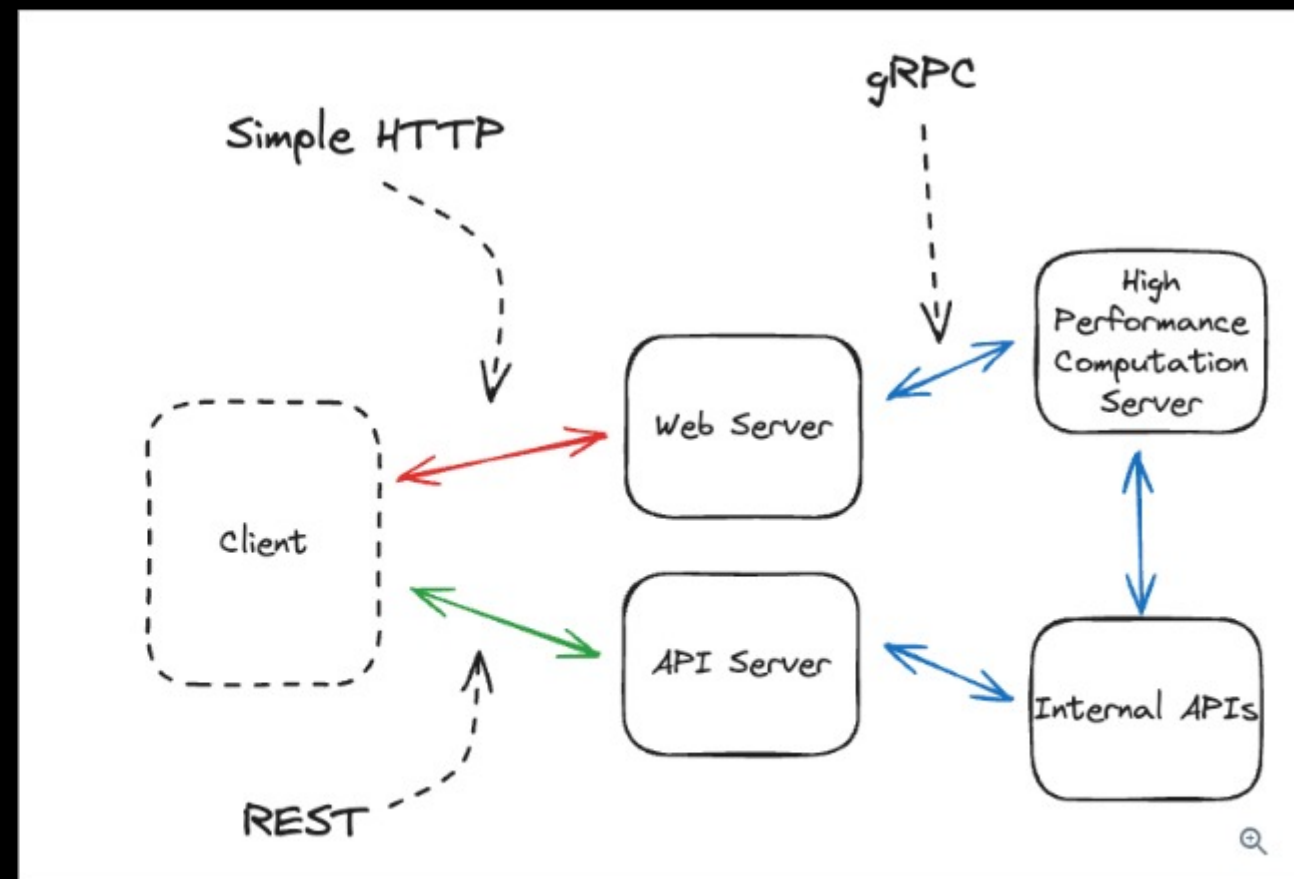
User {
  id
  name
  age
}
    
```

- JSON: 40 bytes

- Serialized: 15 bytes

Query param

?page=4&offset=10



Server Side Events (SSE) - push based

Websockets

- Bidirectional communication with high frequency updates with low latency
- think in terms of messages exchanged instead of req-resp

WebRTC (UDP)

- peer to peer communication
- collaborative document editing, audio-video calls

Scaling

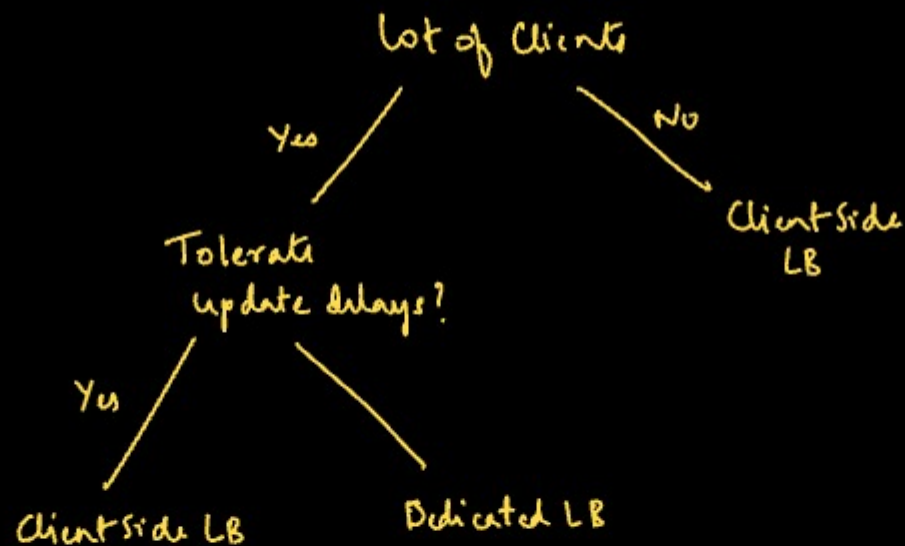
- Vertical scaling
- Horizontal scaling

Load Balancer

- Client load balancer
- A dedicated load balancer

Eg: Redis cluster
DNS

↓
To prevent SPOF
in a LB, we can have
multiple dedicated LBs
& have DNS rotate among
the IPs



- Client LB works very well for internal microservices
- built into gRPC

- Dedicated load balancer

- If we don't want our clients to refresh the list of servers

or

If we don't want to give extra info about servers, use a dedicated LB

- Extra network hop

but we get to decide how to route

- Can occur at two levels

- L₄ load balancers (Transport layer)

- L₇ load balancer (Application layer)



- examines actual data & intelligent routing decisions

- LB is end point for first TCP connection

- make decisions based method, headers, cookies, URL etc

- CPU intensive (examine packets)

- flexible

- suited for HTTP traffic

- Doesn't look inside packets (application data)

- It's like randomly selecting a server

- Works as if there is a direct TCP connection between client & server

- fast & efficient

- raw performance is priority

- use when persistent connection is required

- Eg: websockets

• Health checks & fault tolerance

- LB can do health checks periodically to figure out which servers are up

- Health checks can be configured



TCP request
(200 ⇒ OK
5xx ⇒ down)

• Load balancing algo:

- Round robin
- Randomly decide
- Least connections → useful for persistent connections to prevent accumulation on one server
- Least response time
- IP hash → stateful applications

• Dedicated LB

- hardware LBs — physical devices — can handle 100s of millions req per sec
- software LBs — HAProxy, Nginx etc — order of millions req per sec
- cloud LBs — AWS, GCP

• Regionalization & latency

- Light in fibre optic cables travels at $\frac{2}{3}$ rd of light speed
~ 200 000 km/s

- Distance New York & London — 5600 km between

- Round trip for light itself — $\frac{56 \text{ ms}}{2}$

↓
theoretical
minimum

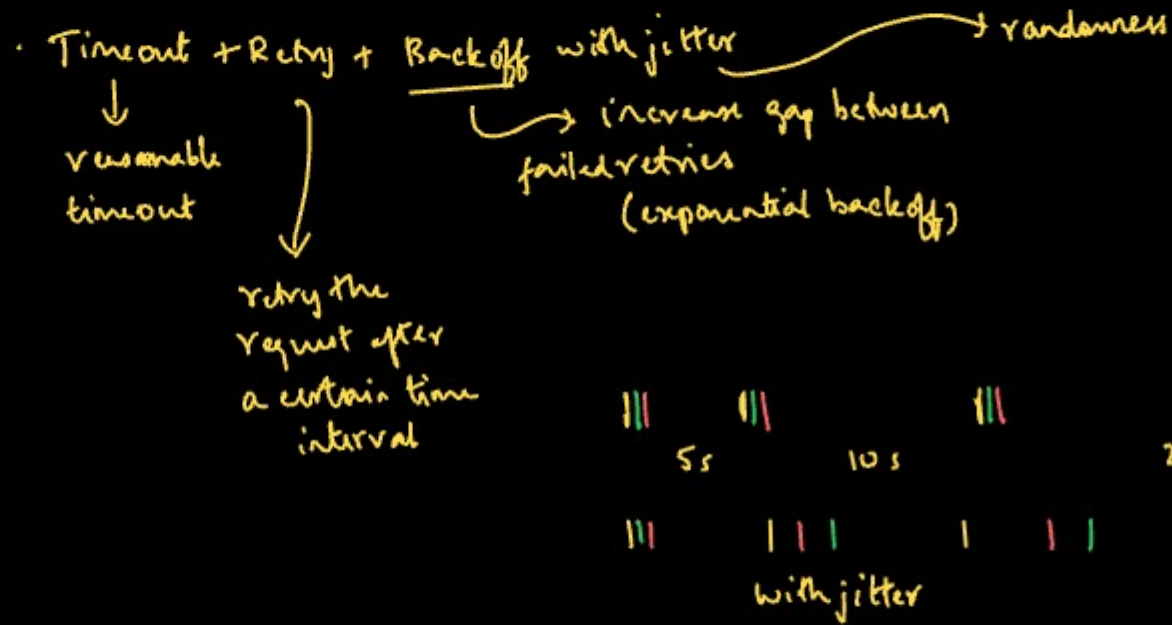
8900 km
⇒ round trip's
theoretical
min is 89 ms

Regionalization

- 1. Keep server & database close
- 2. Keep both of them close to the user
- Have multiple data centers in a single region & repeat this pattern across multiple regions
- CDNs
 - static data & media
 - cached computationally expensive data
- Regional partitioning
 - Uber

Handling failures

- Server failures
- Network failures - Don't assume that network is reliable (dropped packets, corrupted data etc)



- To prevent side effects in retries, we make the requests idempotent
 - Idempotency key: to figure out if the same request

• Circuit Breakers



Cascading failure if DB is down.

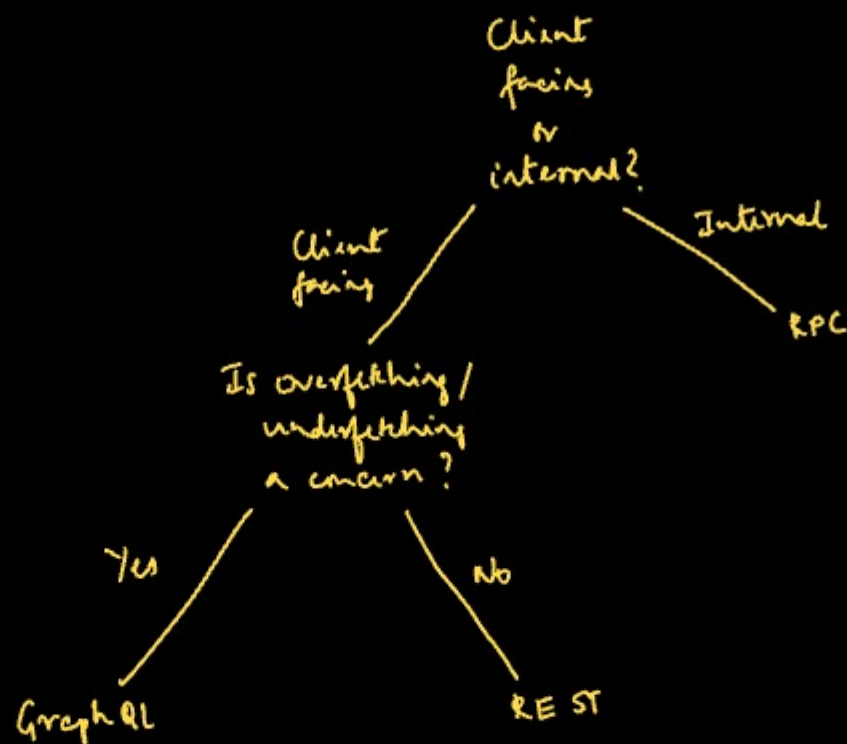
• Circuit breakers

API Design:

- How clients interact with our system

- Three main options

- REST $\rightarrow$ default choice (90% of the time this is the correct choice)
- GraphQL
- RPC



• REST

- Core entities are your resources (nouns)

• Eg: events, tickets, venues, bookings

• Get all events: GET /events

Get a particular event: GET /events/{id}

Create a new booking for an event: POST /events/{id}/bookings

POST /bookings

} plural nouns - always

• Relationships between resources

1. nest resources with path parameters

/events/{id}/tickets



path parameter

2. If we want users to directly interact with tickets - filter, sort etc, we query parameters

/tickets? k₁=v₁ & k₂=v₂ & k₃=v₃

/tickets?eventId=123

• GET - retrieve

(without any side effect)

- idempotent

• POST - create new resources

- not idempotent

• PUT - replace entire resource

- idempotent

• PATCH - update part of a resource

- not idempotent (eg: increment)

• DELETE - removing a resource

- idempotent (first delete - 204
subsequent - 404
deletes)

- Three options to pass data to the API
 - path parameters - eg: resource id
 - query parameters - for optional parameters
 - filters
 - pagination
 - request body - data structures
 - payload

• Return data (response) {

 status code ✓
 response body

 2xx
 4xx
 5xx

• GraphQL

- client decides the shape of data
- use cases:
 - different kinds of devices
 - constantly changing frontend
 - separate independent teams for frontend & backend

• complications

- query parsing, schema validation
- sophisticated caching challenges

• N+1 problem

```

Event {
  venue: Venue!
  ...
}
  
```

- 100 events - get all of them
 - 1 query

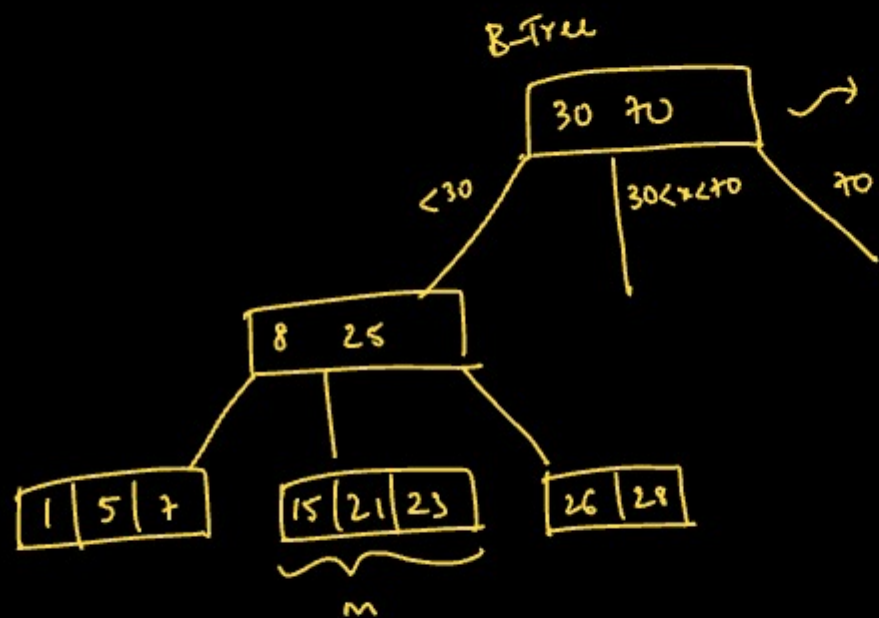
SELECT * FROM events LIMIT 100

- 100 venues' data - 100 queries for each of the 100 venue ids

100 + 1 = 101 queries

- batching / data loader strategies to handle this

- Sharding
 - Break down the DB into multiple parts (shards)
 - Each shard is its own DB with its own CPU, memory etc



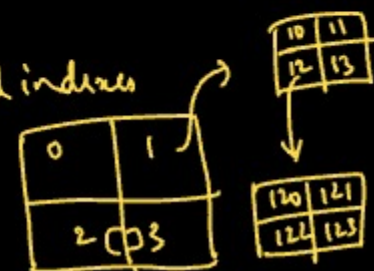
Each node fits in a page (4KB)

- 1 page - 1000 rows
- 10 million rows in user table
- 10000 pages
- Index:
 - Hash based Index - exact match in $O(1)$ on avg
 - B-Tree Index

- BST
 - 10 million nodes
 - Each node - 100 bytes
 - Total memory required = 10^9 Bytes = 1GB

PostgreSQL

- default - index on primary key, unique constraints
- we manually add index based on columns of frequently accessed queries
- WHERE email LIKE "%abc%"
- Full text search (Inverted index)
 - word₁ → [row₁, row₂]
 - word₂ → [row₂, row₃, row₅]
 - ⋮
- For location based queries: Geospatial indexes
 - Geohash
 - Quadtree
 - R-Tree



Two decisions

1. What to shard by?
2. How to distribute across shards?

E-commerce

- created-at field of orders - shard key X
- hot shards - focused on the latest orders
- is-premium-user - low cardinality X

Sharding strategies:

Hash based (with consistent hashing) - default

Simple modulo based hashing

- 4 shards



5 shards

Approximately how much data is moved?

- u_1 - user id₁
- u_2 - user id₂

$$h(u_1) \% 4 = h(u_1) \% 5 \Rightarrow \text{NO shifting}$$

$$20n, 20n+1, 20n+2, 20n+3 \quad \checkmark \quad 4/20 \text{ won't shift} \Rightarrow \sim 80\% \text{ shifts}$$

Random shard X

Directory based sharding

- key $\rightarrow$ shard # map
- allows us to have custom logic for each shard key

Cons
1. directory - SPOF
 $\Rightarrow$ extra step to figure which shard # key is in

Three properties:

1. High cardinality
2. Evenly distribute it across shards
 - we don't want to have hot shards
3. Align with queries
 - All the data required for a query should ideally be on one shard

user-id of user table $\rightarrow$ shard key

- generally queries in a social media app are user centric

- all posts, likes, comments of a user go into the same shard.

Range based sharding

- 0-1M $\rightarrow$ shard 1
- 1-2M $\rightarrow$ shard 2

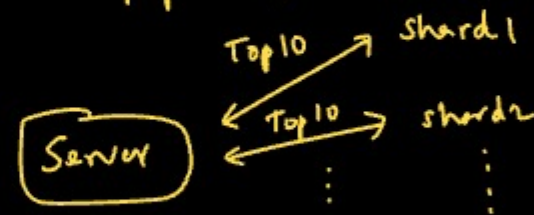
- might not create even distribution
- Each company in a SAAS app has its own range of IPs
 - works perfectly

Challenges:

- Hot keys - Celebrity problem
 - Isolate all hot keys to a dedicated shard
 - directory based sharding
- Compound shard key
 - (user-id, date) → shard key combination
 - distributes it across multiple shards
- Dynamic shard splitting:
 - MongoDB, Vitess
 - DB provider automatically detects it & breaks it down into further shards
 - not operator driven

Cross shard queries:

- Always going to be there
- Want to minimize
- Eg: Top 10 most popular posts



Aggregate,
sort & get
top 10.

Strategies:

- Precompute,
 - Caching the computed result
 - TTL - 60s
 - 55s (about to expire) - background job to precompute.
- Denormalization
 - duplicating data (updates are more complex)
 - avoid cross-shard queries
- Accept it sometimes

Consistency

- Single DB - transactions are straightforward
- Many issues with distributed transactions
 - machine failures, network failure, synchronizing time, DB updated in between individual queries of transaction etc
- 2 phase commit:
 1. Coordinator asks shards to prepare for transaction
 2. Once all acknowledgements are received, asks everyone to commit
- slow & fragile → during this process if any of the shards/coordinator fails, the system gets stuck
- most production systems avoid this due to these tradeoffs.

1. Try to avoid cross-shard queries
2. Sagas - for cross shard pattern
 - each step - has a corresponding undo step
 - If any steps of transaction fail, we undo the previous steps in reverse order
 - eventual consistency without the fragility of 2PC
3. Accept eventual consistency
 - Eventual consistency is acceptable for a lot of systems.
 - Eg: user follower count
 - We'd have a brief period of inconsistency

- Cassandra, DynamoDB, MongoDB
 - we just need to specify the sharding key & they handle the rest
 - automatically rebalance when necessary
 - Vitess & Citus — act as sharding layers that sit in front of SQL DBs (PostgreSQL, MySQL)

operator driven
resharding

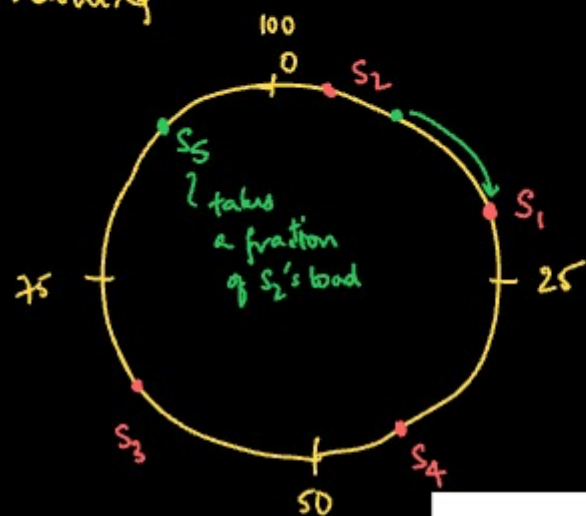
- AWS Aurora, Google Cloud Spanner
 - distributed SQL with inbuilt

- Avoid premature sharding
 - establish (with numbers) why we need sharding
- Sharding — to increase storage & throughput
 - 2 decisions
 - 1. shard key
 - 2. distribution strategy
 - justify sharding
 - shard key choice
 - explain trade offs
 - show that you have thought about resharding & cross shard queries

Consistent Hashing

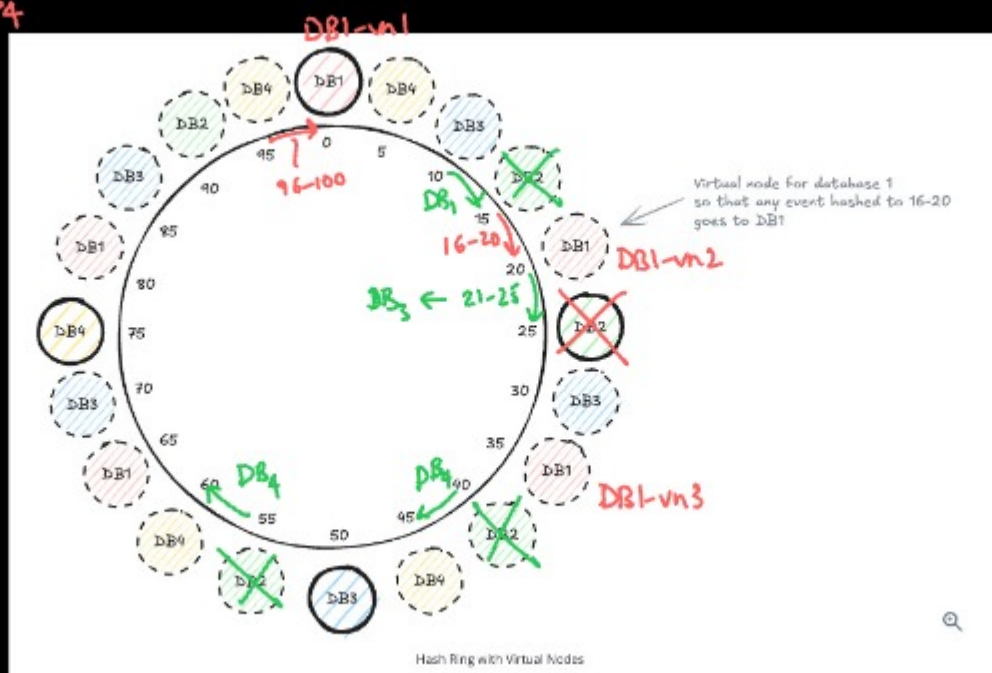
- Simple modulo based hashing-downside
 - Add/delete a shard
 - $\Rightarrow$ lot of data needs to be remapped

Consistent hashing



Virtual nodes:

- $h("DB1") \% 100$
- $h("DB1-vn1") \% 100$
- $h("DB1-vn2") \% 100$
- $h("DB1-vn3") \% 100$
- $\vdots$
- $h("DB1-vn5") \% 100$



- Uniform distribution: all values in the range are equally likely



- Normal distribution



- Can hotspots still occur?

- Yes
- CH (even with V-nodes) doesn't solve this problem

• Strategies for hotspot issue

• Read replicas

- replicate data across multiple shards
- & use some load balancer

• Key space salting:

- Cryptography: We add a small randomized string to our data

- $usr1 \rightarrow \text{hot key}$

$\left. \begin{array}{l} \text{usr1-0} \\ \text{usr1-1} \\ \text{usr1-2} \\ \vdots \\ \text{usr1-9} \end{array} \right\} \text{shard keys}$

put $usr1$ data
in all shards
these salted keys
hash to.

• Adaptive rebalancing

- Some DB providers automatically reshard

• replication

- copies of data on multiple machines
- leader-follower & multi leader replication

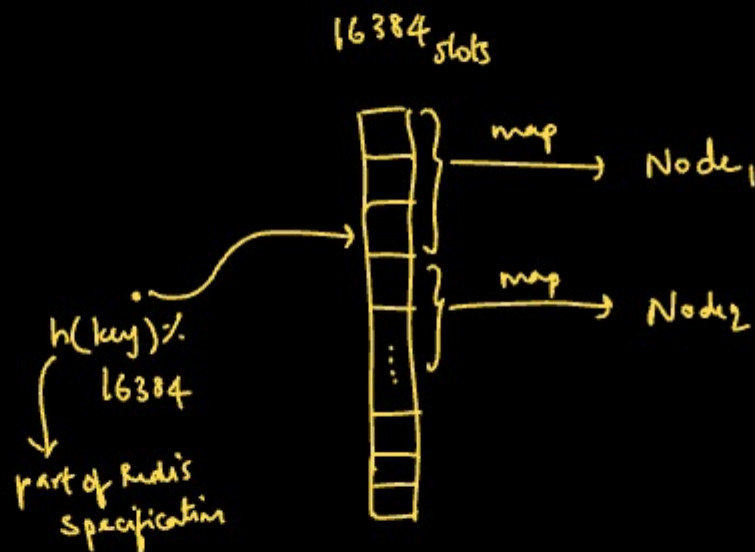
• CH

- Caches, message-brokers, application server

- Cassandra
- DynamoDB
- CDNs

• Redis - doesn't use CH

- Fixed hash slot approach - Redis clusters



slots to nodes
mapping can
change dynamically

• gossip protocol

↓
Redis uses this to communicate
mapping info across nodes

- every node has the mapping info.